

MACHILIPATNAM, India

WMO: 431850

Lat: 16.1829N

Lon: 81.1533E

Elev: 10

StdP: 14.69

Time Zone: 5.50 (IND)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	67.0	68.3	59.3	75.3	80.1	61.5	81.6	78.7	12.3	79.1	11.0	79.3	2.2	0	0.331

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	15.4	105.0	80.0	100.7	80.4	97.9	80.8	84.9	94.7	84.1	93.2	83.4	92.1	5.7	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
82.7	170.4	90.2	81.9	166.1	89.3	81.3	162.5	88.7	49.4	95.5	48.4	93.6	47.6	92.0	88.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 13.8	11.7	9.9	62.3	111.5	3.9	3.5	59.5	114.0	57.3	116.1	55.1	118.0	52.3	120.6		
(5)			60.5	86.6	1.6	1.1	59.4	87.4	58.4	88.1	57.5	88.7	56.4	89.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	83.8	76.7	79.4	83.5	87.5	91.3	89.5	86.3	85.3	85.2	83.3	80.2	77.1
	DBStd	5.39	1.93	2.08	2.16	2.31	4.01	4.66	3.76	3.26	2.80	2.71	2.40	2.21
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD50	12335	828	824	1037	1125	1280	1185	1124	1096	1057	1034	905	840
	CDD65	6860	363	404	572	675	815	735	659	631	607	569	455	375
	CDH74	81769	2410	3496	6533	9112	12047	10520	8570	7947	7636	6555	4279	2664
	CDH80	39155	546	1181	2822	4886	7611	6289	4288	3698	3472	2516	1243	603
Wind	WSAvg	3.9	3.0	3.2	3.8	5.2	5.2	4.9	4.2	3.9	3.0	2.9	3.5	3.4
Precipitation	PrecAvg	42.8	0.2	0.5	0.3	0.4	1.5	4.5	7.2	6.7	7.0	8.3	4.7	0.8
	PrecMax	82.4	1.9	3.7	5.3	7.0	9.2	13.0	17.7	16.4	21.4	30.1	21.5	8.9
	PrecMin	19.6	0.0	0.0	0.0	0.0	0.0	0.6	1.8	0.8	0.4	0.7	0.0	0.0
	PrecStd	12.0	0.4	1.0	1.0	1.1	2.0	3.0	3.5	3.5	4.3	6.1	5.2	1.6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	88.0	92.6	95.6	99.0	112.0	108.4	100.8	98.2	96.8	94.8	91.1	88.3
		MCWB	74.5	75.6	79.2	82.0	77.5	80.6	78.6	79.7	80.2	79.8	77.1	74.6
	2%	DB	85.9	90.0	93.5	97.0	107.5	104.0	98.0	95.3	94.7	92.6	88.8	86.1
		MCWB	73.6	75.4	78.9	82.0	78.6	80.2	78.7	79.7	80.4	79.9	76.7	74.1
	5%	DB	84.5	88.2	91.9	95.5	102.4	100.5	95.5	93.3	92.8	90.8	87.2	84.7
		MCWB	72.9	75.2	78.6	81.7	80.4	80.2	79.2	80.0	80.5	79.8	76.4	73.3
	10%	DB	83.2	86.4	90.5	94.1	98.3	97.4	93.0	91.5	91.0	89.0	85.9	83.4
		MCWB	72.5	74.8	78.2	81.3	82.2	80.0	79.3	79.8	80.5	79.5	76.1	72.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	77.4	80.0	82.0	84.6	86.5	85.6	83.4	83.2	83.4	82.9	80.6	77.4
		MCDB	84.4	88.2	91.7	94.4	97.3	96.6	91.9	90.9	90.9	90.0	86.9	84.1
	2%	WB	75.7	78.1	81.0	83.7	85.3	84.4	82.3	82.1	82.5	81.9	79.5	76.5
		MCDB	82.4	86.3	90.1	93.0	95.3	94.3	90.5	89.6	89.8	88.9	85.5	82.9
	5%	WB	74.6	76.9	80.2	82.9	84.5	83.4	81.6	81.5	81.8	81.2	78.7	75.6
		MCDB	81.4	84.8	88.8	91.9	93.8	92.6	89.2	88.7	88.7	87.7	84.5	81.7
	10%	WB	73.7	75.9	79.3	82.2	83.8	82.5	80.9	80.9	81.2	80.5	77.9	74.7
		MCDB	80.6	83.4	87.4	90.9	92.5	91.1	88.0	87.8	87.9	86.7	83.3	80.6
Mean Daily Temperature Range	5% DB	MDBR	14.5	15.5	15.5	14.5	15.4	14.1	11.9	11.1	10.9	10.8	12.3	14.1
		MCDBR	15.7	18.0	17.0	15.6	22.2	18.6	15.4	13.9	13.6	13.6	13.8	15.1
		MCWBR	5.7	6.2	5.4	4.4	6.5	5.6	4.9	4.5	4.4	4.9	5.1	5.6
	5% WB	MCDBR	14.0	15.5	15.2	14.4	15.4	15.2	13.1	12.2	11.8	11.3	10.9	12.8
		MCWBR	5.8	6.3	5.5	4.7	5.5	5.7	5.2	5.0	4.9	4.9	4.8	5.8
Clear-Sky Solar Irradiance	taub	0.522	0.509	0.563	0.578	0.651	0.620	0.573	0.559	0.553	0.563	0.544	0.564	
	taud	1.975	2.030	1.923	1.928	1.779	1.859	1.989	2.028	2.041	2.000	1.987	1.900	
	Ebn at Noon	243	254	245	240	220	225	237	242	244	237	235	226	
	Edh at Noon	56	55	62	62	71	65	57	56	55	56	54	58	
All-Sky Solar Radiation	RadAvg	1529	1788	1955	2091	1945	1556	1422	1463	1505	1479	1435	1390	
	RadStd	56	67	73	76	69	131	168	119	124	136	103	89	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		+0.54	+0.58	N/A	N/A	+0.56	+0.54	N/A	N/A	N/A
(47) Regional (3 neighbors)		+0.54	+0.58	+0.86	N/A	+0.36	N/A	N/A	N/A	N/A

Nomenclature: See separate page